

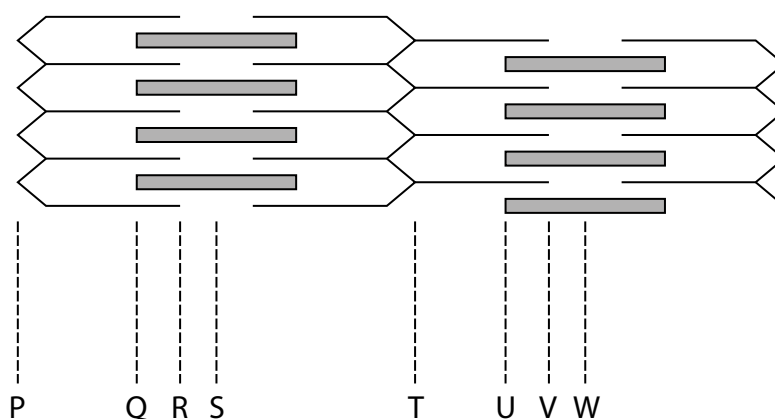
4 Movement in mammals results from the interaction between muscles, tendons, ligaments and the skeleton.

(a) Which row of the table describes tendons and ligaments?

(1)

	Tendons	Ligaments
☐ A	flexible, non-elastic tissue connecting bone to bone	flexible tissue connecting muscle to bone
☐ B	flexible, non-elastic tissue connecting muscle to bone	flexible tissue connecting bone to bone
☐ C	flexible tissue connecting bone to bone	flexible, non-elastic tissue connecting muscle to bone
☐ D	flexible tissue connecting muscle to bone	flexible, non-elastic tissue connecting bone to bone

(b) (i) The diagram shows part of a skeletal muscle fibre.



Which pair of letters represents one sarcomere?

(1)

- ☐ **A** P to T
- ☐ **B** Q to U
- ☐ **C** R to V
- ☐ **D** S to W

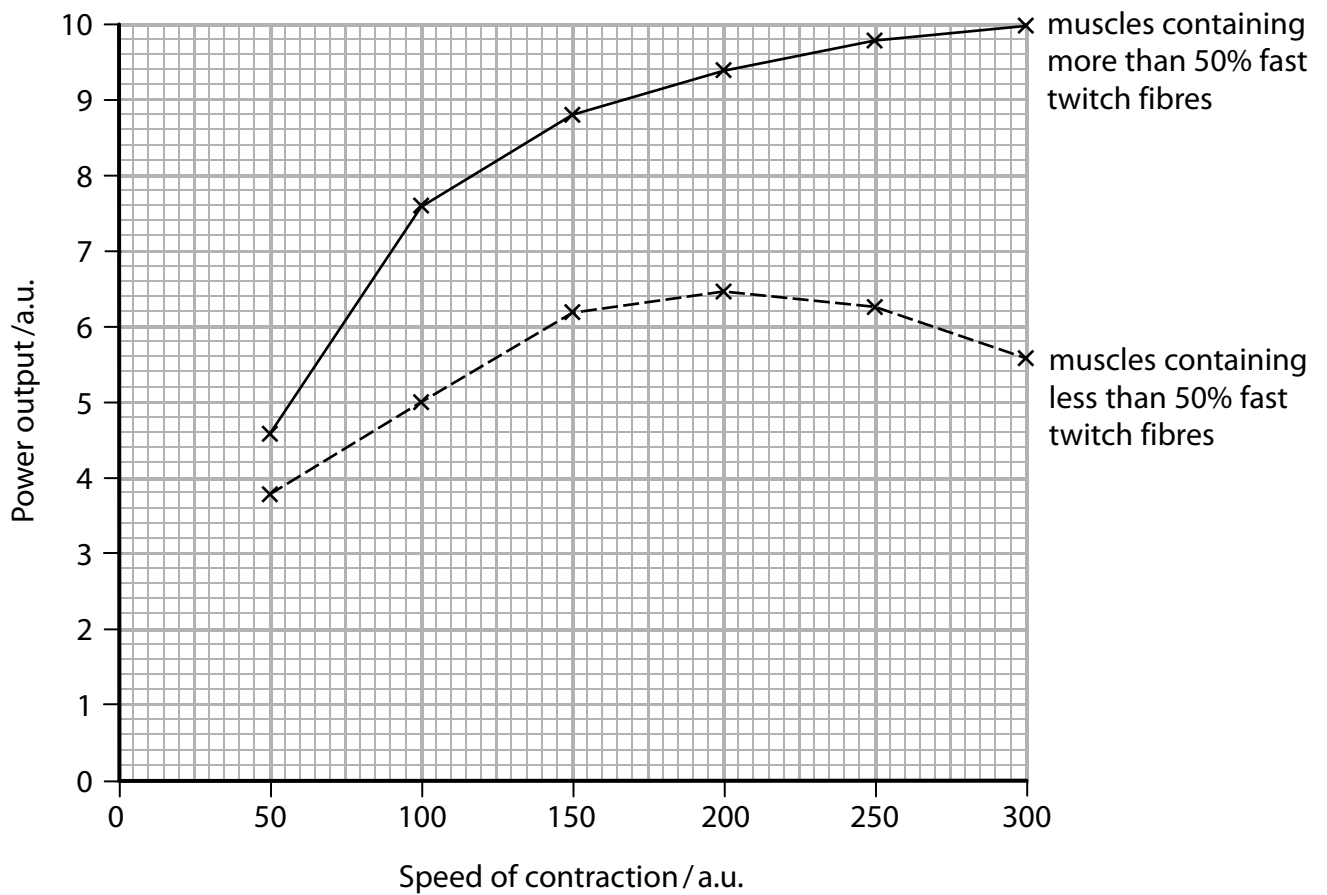
(ii) Compare and contrast the structure of fast-twitch and slow-twitch muscle fibres.

(4)

[illegible]

*(c) The power output of a muscle depends on the speed of contraction and the proportion of fast-twitch and slow-twitch fibres in the muscle.

The graph shows the relationship between the power output and the speed of contraction of two different types of muscle.



Explain the relationship between power output and the speed of contraction of these two types of muscle. Use the information in the graph and your own knowledge to support your answer.

(6)

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(Total for Question 4 = 12 marks)